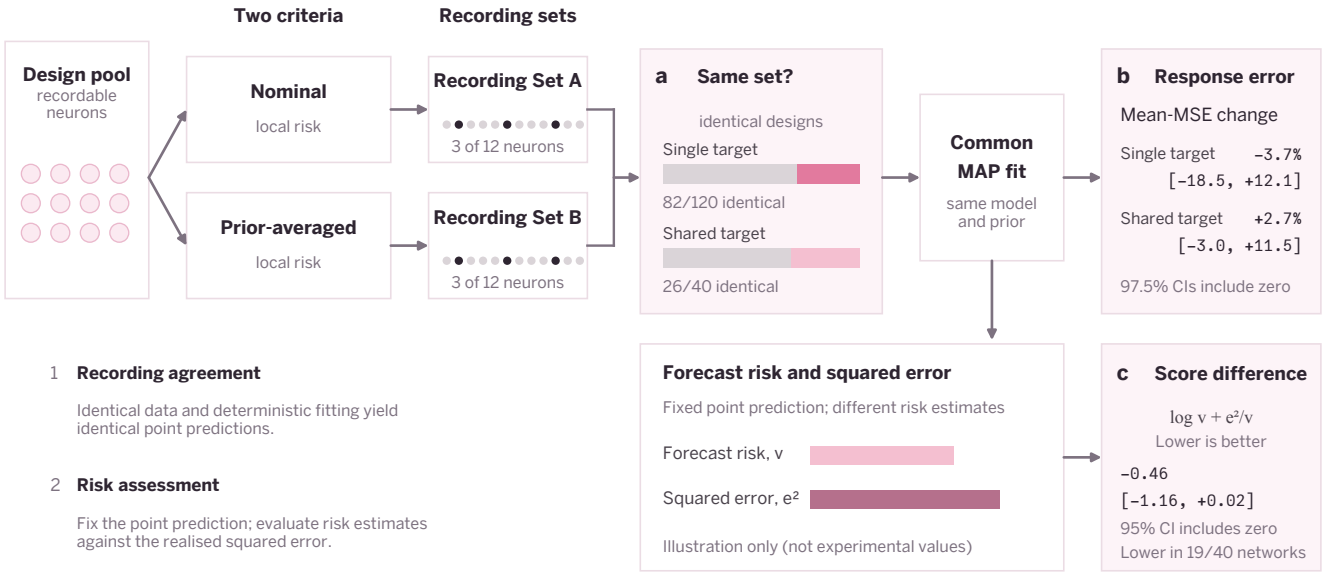
Fig. 1 | Three-layer audit



1 **Recording agreement**

Identical data and deterministic fitting yield identical point predictions.

2 **Risk assessment**

Fix the point prediction; evaluate risk estimates against the realised squared error.

Fig. 2 | Connectivity, recordings and response



Fig. 3 | Recording decisions and response error

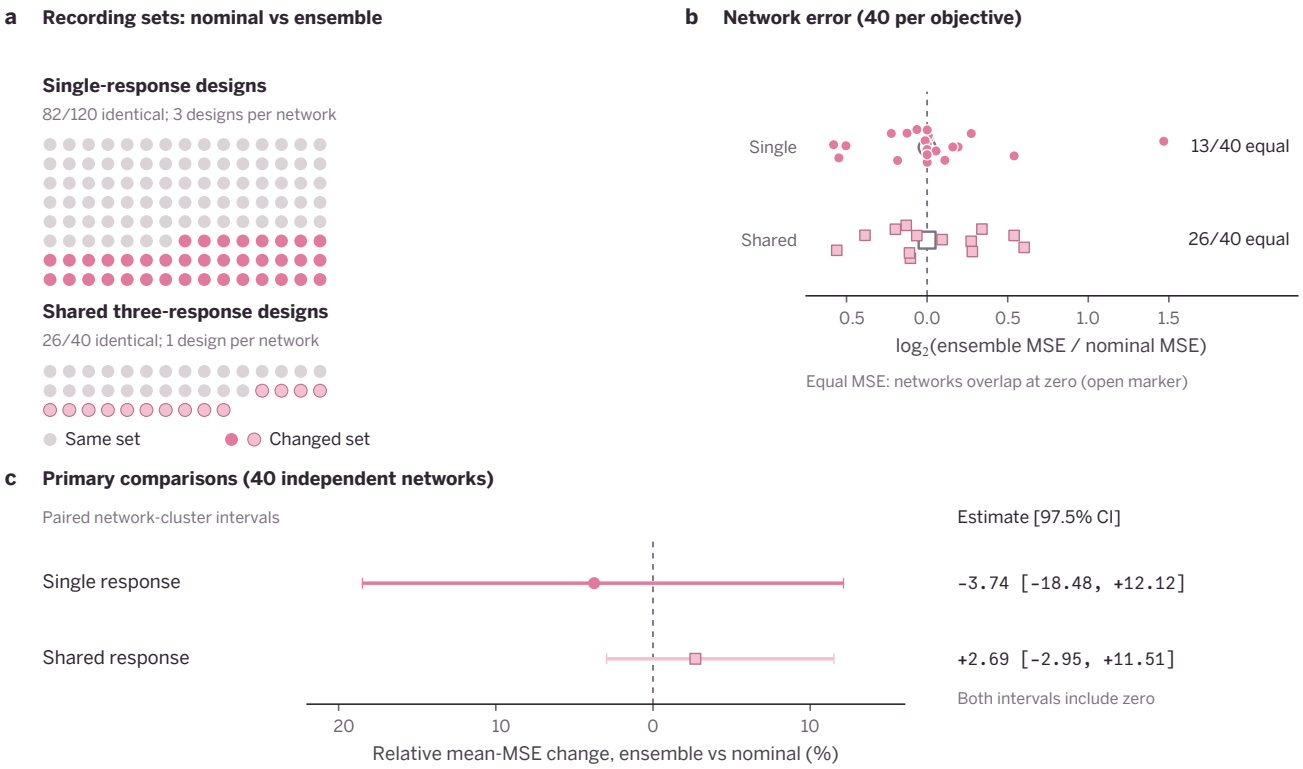
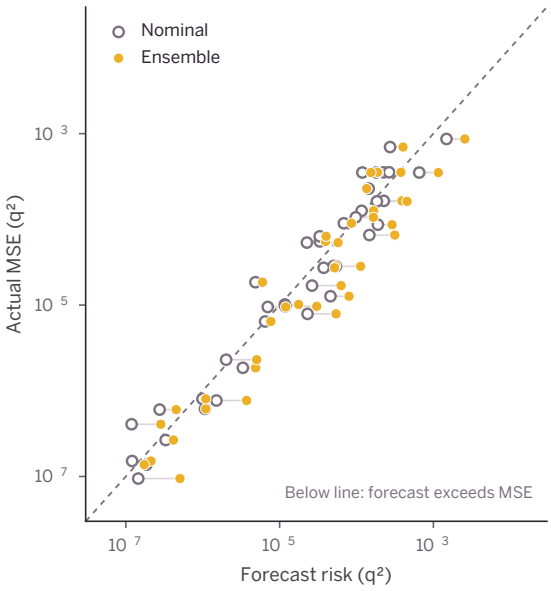
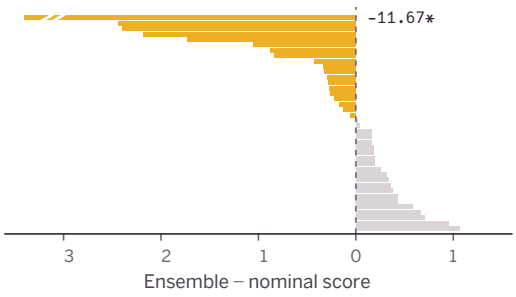


Fig. 4 | Risk forecasts on fixed predictions

a **Forecast risk and actual MSE**



b **Network-level score differences**



40 networks: 19 lower; 21 higher. *Clipped at 3.4.
One network: 64% of net summed score difference.

c **Mean score difference**

-0.46 [-1.16, +0.02], 95% network-cluster CI

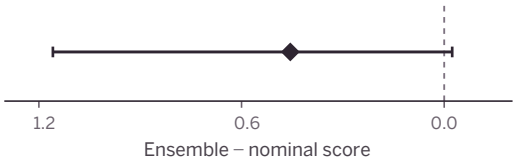
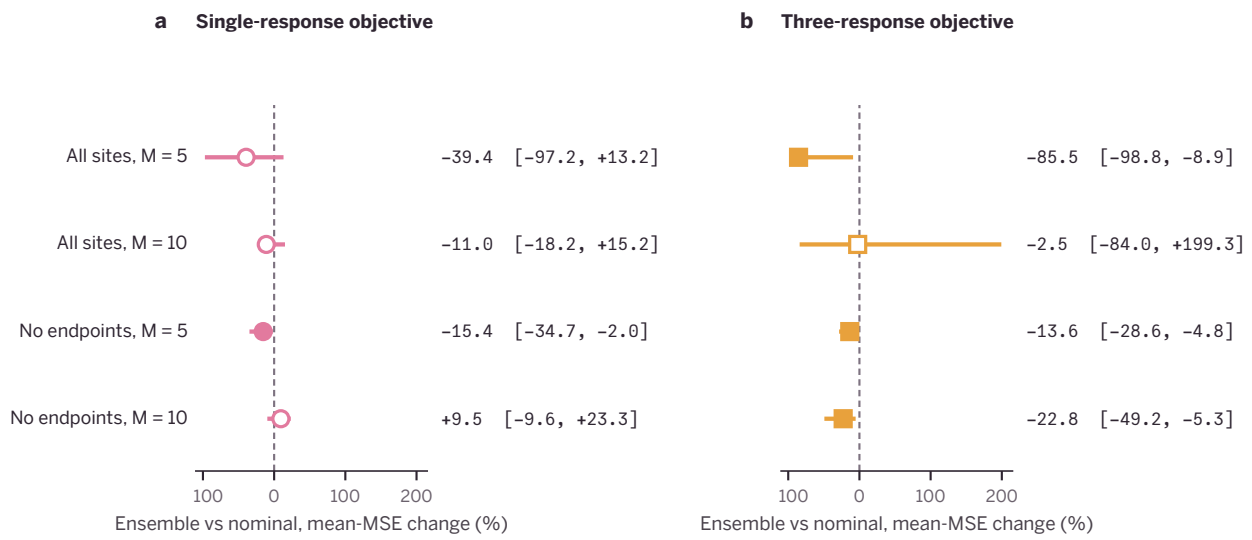


Fig. 5 | Exploratory model-family extension



Filled marker: 95% interval excludes zero (4 of 8 settings, all favouring the ensemble). Hollow: includes zero.
Exploratory: 8 independent 100-unit networks, pointwise intervals, no multiplicity adjustment; model family and N change together.

Fig. S1 | Paired distributions

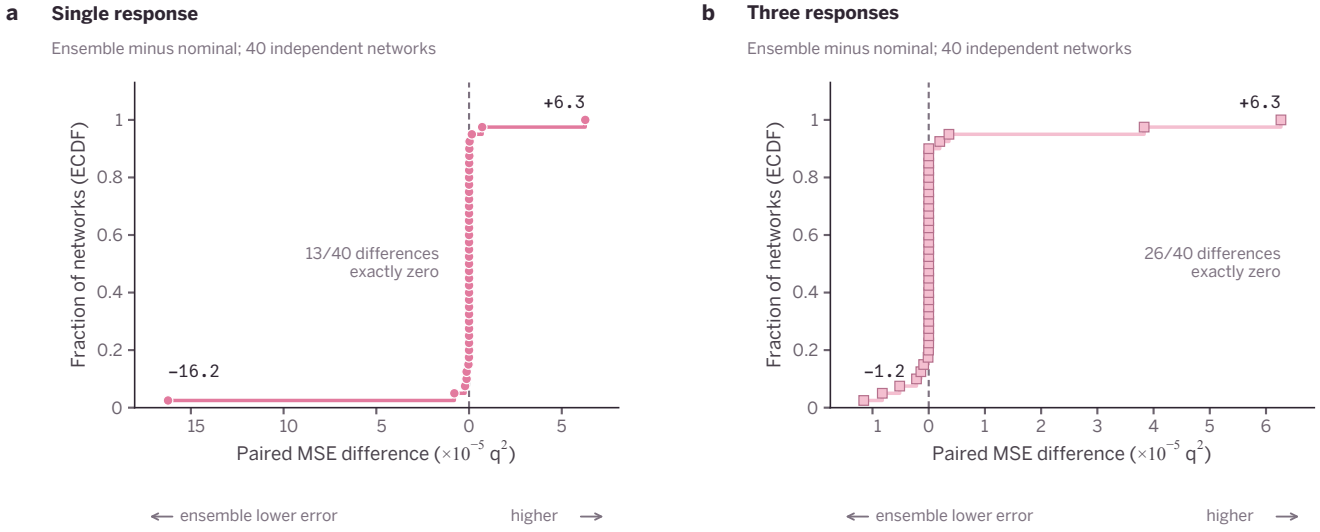
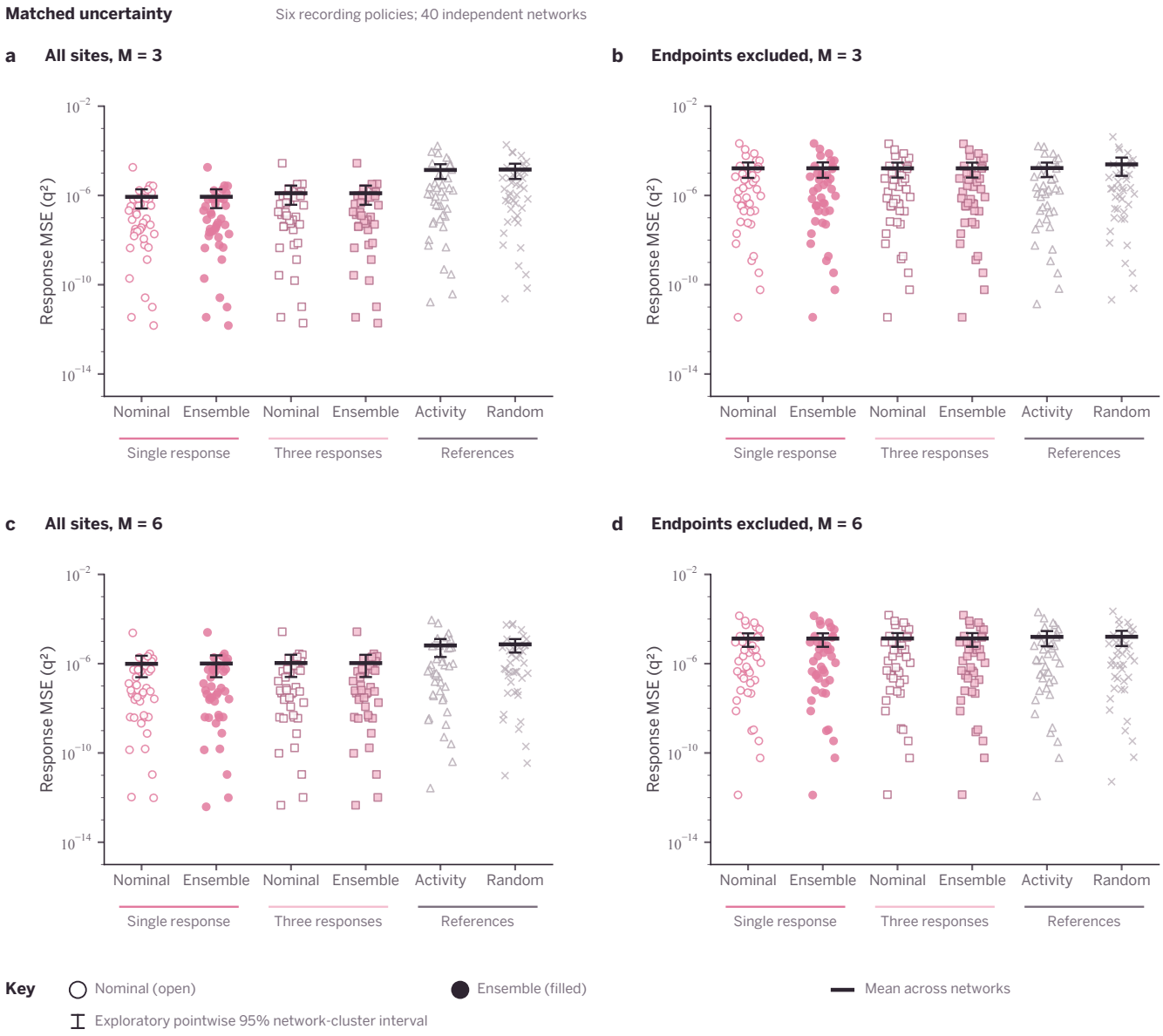
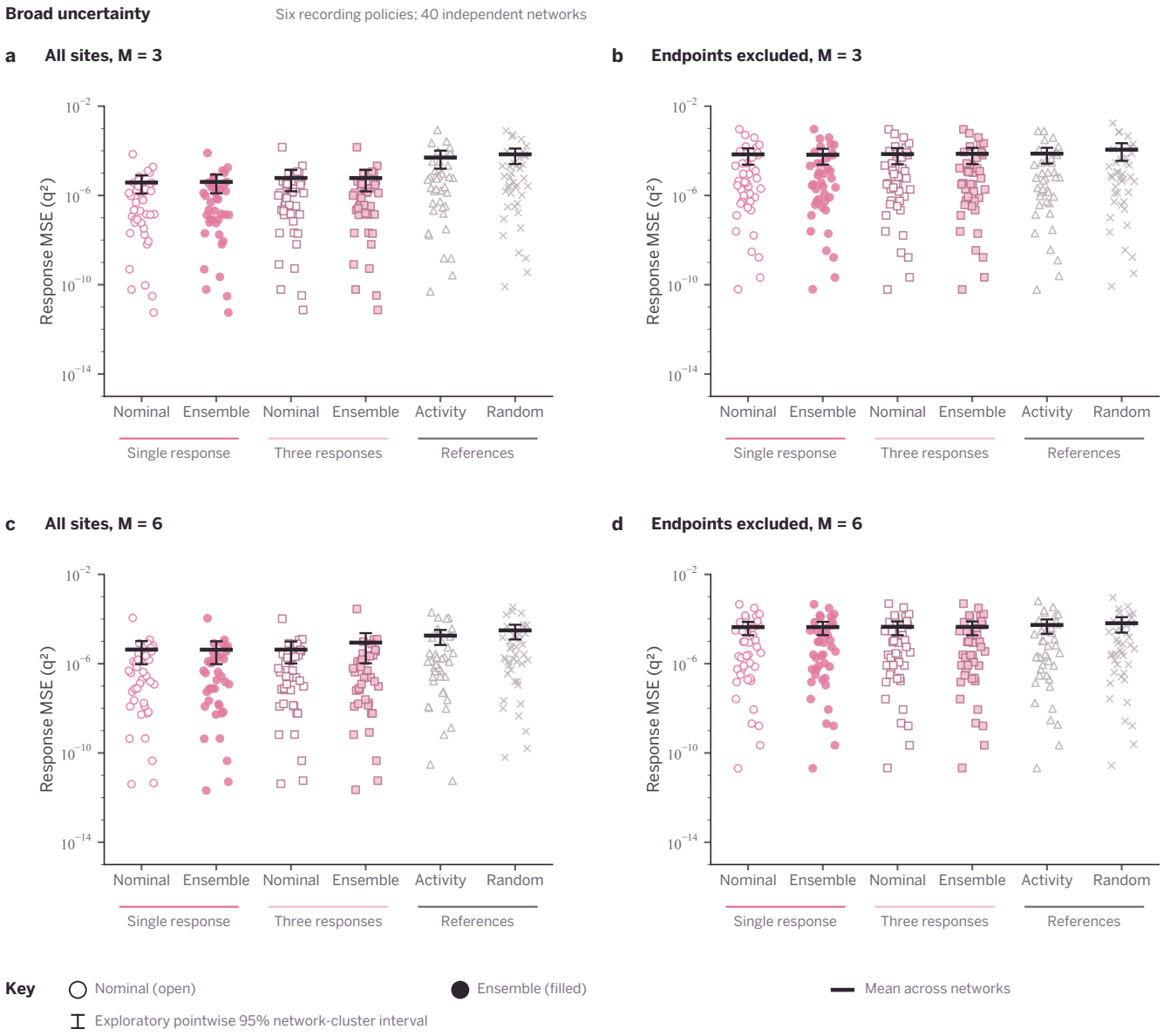


Fig. S2 | Matched uncertainty



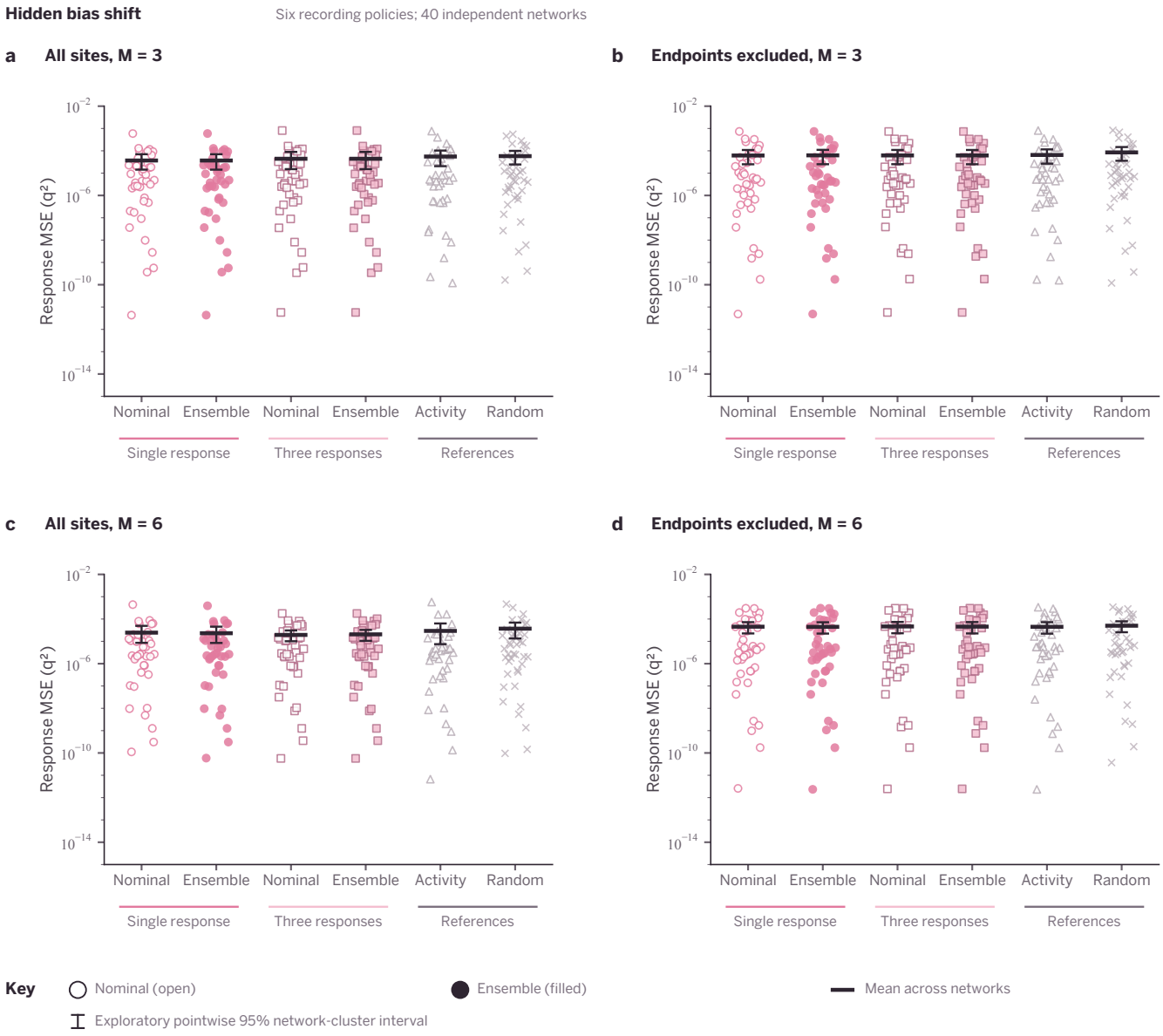
Circle: single response; square: three responses; triangle/cross: references.
Each point is one network, averaged over truth, noise and readout repeats. All 40 networks shown; horizontal jitter only.

Fig. S3 | Broad uncertainty



Circle: single response; square: three responses; triangle/cross: references.
Each point is one network, averaged over truth, noise and readout repeats. All 40 networks shown; horizontal jitter only.

Fig. S4 | Bias-shift uncertainty

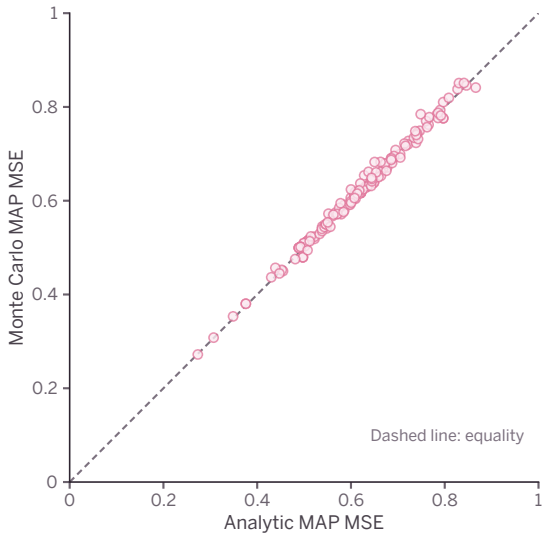


Circle: single response; square: three responses; triangle/cross: references.
Each point is one network, averaged over truth, noise and readout repeats. All 40 networks shown; horizontal jitter only.

Fig. S5 | Linear positive control

a **Analytic versus Monte Carlo MAP MSE**

Linear-Gaussian positive control, 12 independent systems



b **Linear-Gaussian control**

Independent systems	12
Target–design evaluations	144
Monte Carlo draws per system	8,192

Numerical agreement

Checks the linear-Gaussian risk calculation and MAP fit; biological validity is not tested.

The 144 evaluations are nested within 12 systems. No confidence intervals are shown.